

Analyzing Solar Modulation of Galactic Cosmic Rays Using Fermi-LAT Observations

BACKGROUND

Solar Modulation: The process by which the Sun's magnetic field influences the intensity of galactic cosmic rays (GCRs) in our solar system, creating an 11-year solar cycle. *High solar activity (solar maximum) reduces GCR flux near Earth, and low solar activity (solar minimum) increases it.*

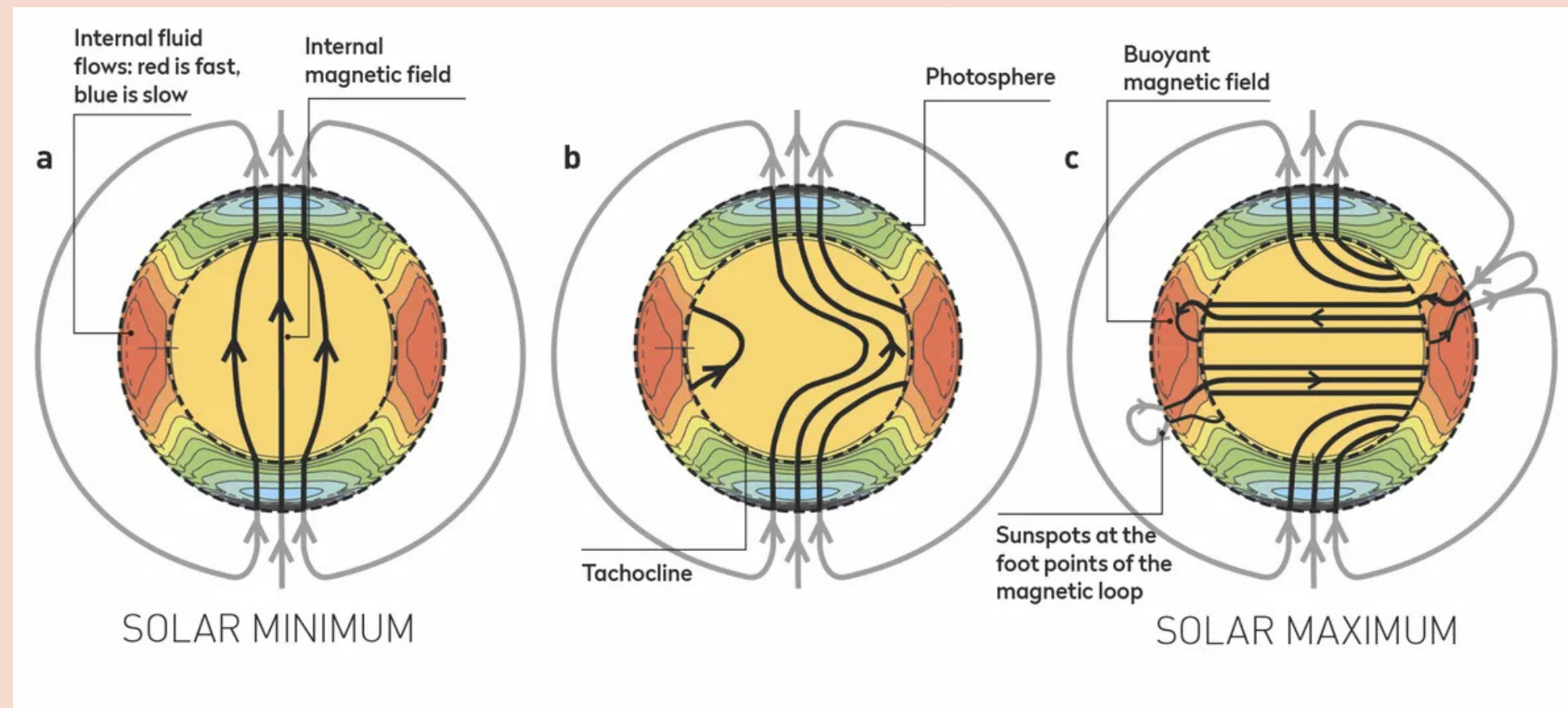


Figure 1: Representation of Sun's internal mechanisms during solar min. and max. Source: BBC

Galactic Cosmic Rays (GCRs): Energetic particles originating from outside of our solar system, usually from supernovae, that constantly bombard Earth. Radiation emitted from these particles can be detrimental to humans and electronics that are outside of Earth's magnetic field.

The Sun's magnetic field (heliosphere) shields the solar system, with **more GCRs penetrating during solar minimum and fewer during solar maximum.**

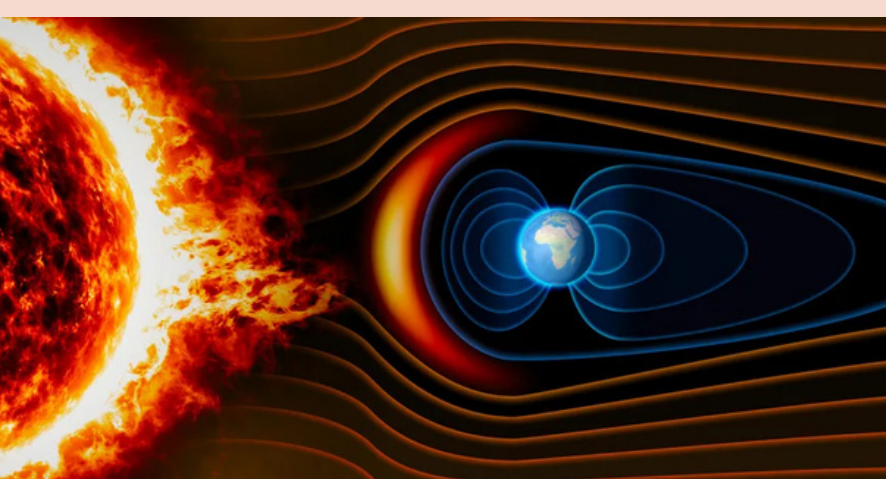


Figure 2: Visualization of Earth's magnetic field deflecting solar activity. Source: NASA

Solar Minimum = less GCRs blocked

Solar Maximum = more GCRs blocked

Fermi Gamma Ray Space Telescope: Able to detect Galactic Cosmic Ray (GCR) flux (the flow of such particles), specifically gamma-rays, near the Earth's surface.

Figure 3: Fermi Gamma Ray Space Telescope. Source: NASA

SIGNIFICANCE

Space Weather: Understanding how solar activity protects our vital infrastructure, like power grids and GPS systems.

Future Space Missions: GCRs ionize everything in their way, including astronauts and electronics, including satellites and spacecraft.

HYPOTHESIS

If solar magnetic activity **increases** during solar maximum, then gamma-ray flux near Earth will **decrease** due to enhanced shielding of GCRs by the heliospheric magnetic field.

Additionally, speed will be **lower** at higher gamma-ray flux as the Sun will be at solar minimum.

METHODOLOGY

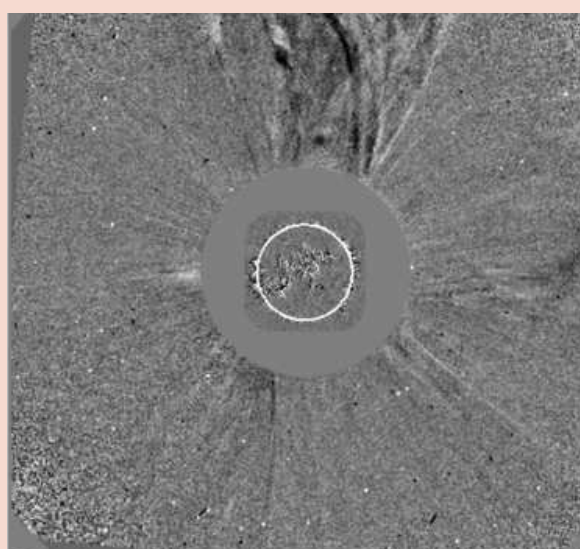


Figure 4: CME visual data from 11/09/2025. Source: NASA

Statistical Analysis

Spearman Correlation
Solar activity vs.
gamma-ray flux

Cross-Reference

CMEs via NASA SOHO
LASCO Catalog
Matched dates with
gamma-ray flux data

Process Data

Gamma-ray flux via
Fermi-LAT
Solar Cycle 25
(Min: Dec 2019;
Max: Oct 2024)

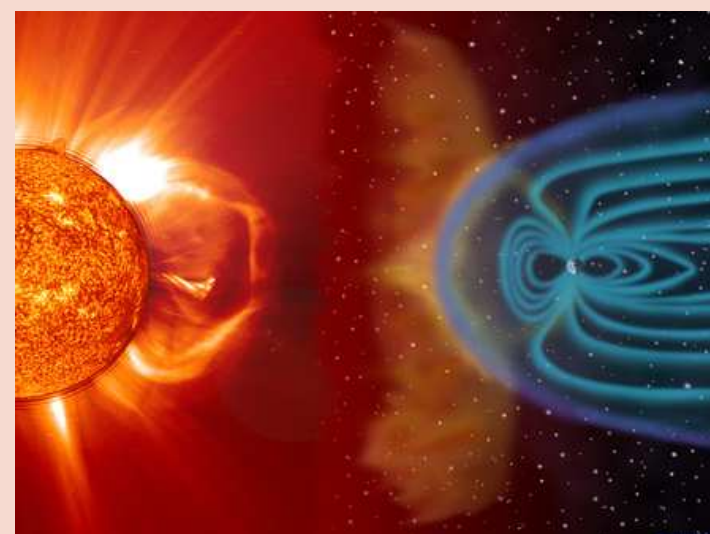


Figure 6: The Sun and Earth's magnetic fields. Source: NASA

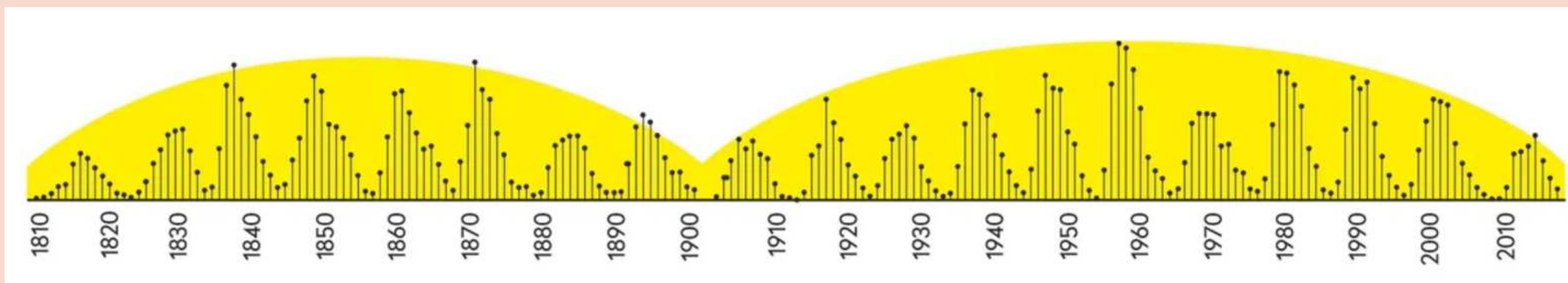


Figure 7: Representation of the 11-year solar cycle over time. Source: BBC

```
from astropy.io import fits
import pandas as pd

filename = "lat_1c_20191214.fits"
date = filename[-12:-8]
hdu = fits.open(filename)
pf = pf.open(hdu[1].data)
pf.read()

for col in pf.columns:
    pf = pf[col].data
    if header.get("datatype") and pf.dtype == "<f":
        pf = pf.astype(float)
    pf = pf.reshape(pf.shape[0])

import matplotlib.pyplot as plt

time = pf["TIME"]
counts = pf["COUNTS"]
counts[where are (len(df["COUNTS"]) - 8)] out of (len(df)) observations that are not pf["COUNTS"]

plt.figure(figsize=(10,6))
plt.plot(counts, label="Counts")
plt.plot(counts, label="Counts (data)")
plt.xlabel("Counts")
plt.ylabel("Counts")
plt.legend()
```

Figure 5: Snippet of the code used

Results

Python
Gamma-ray flux vs. CME
activity

Visualization

Create graphs
Representation of results

RESULTS

Observations:

- Lower gamma-ray flux counts during *solar max*
- Higher gamma-ray flux counts during *solar min*

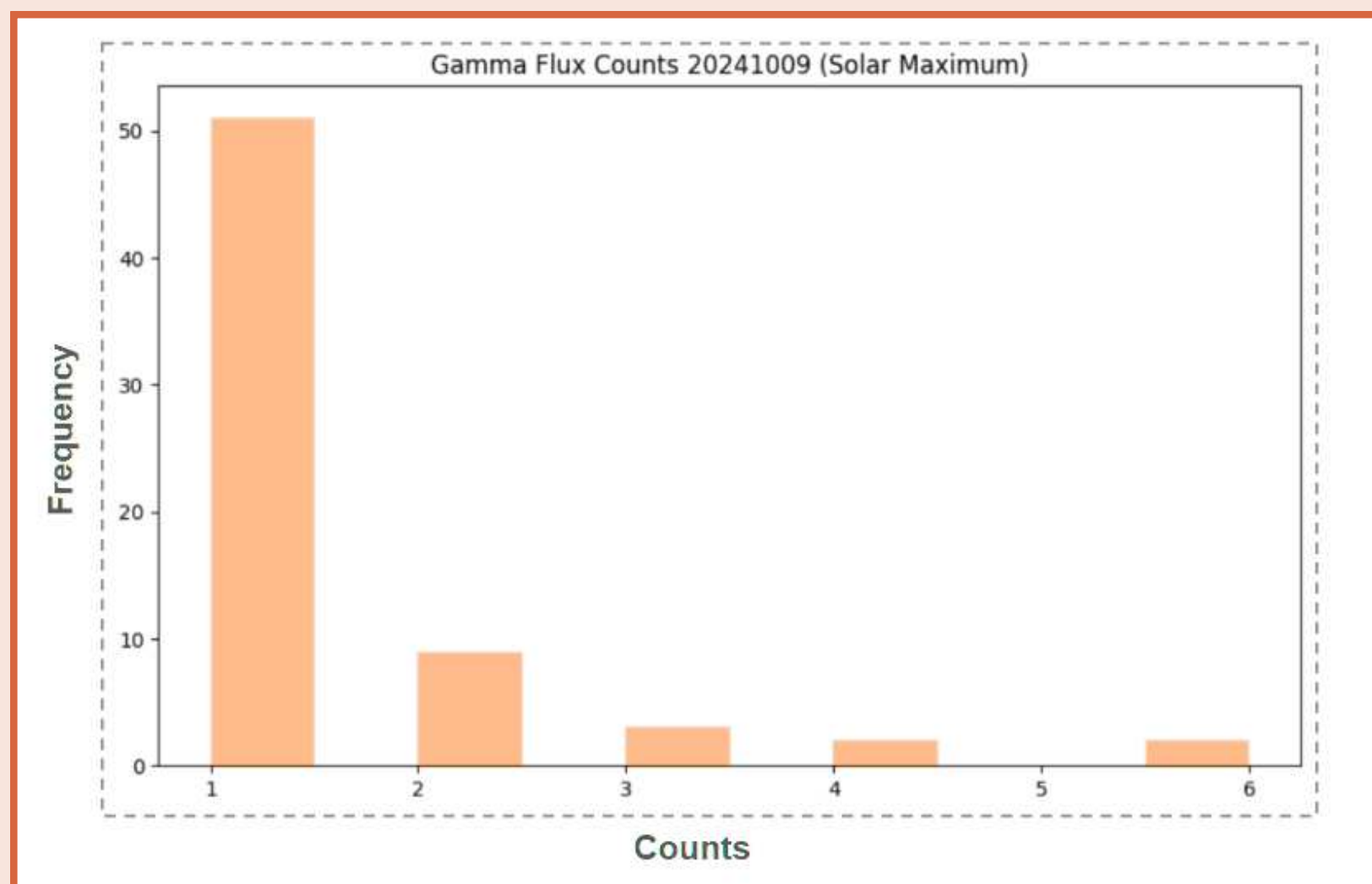


Figure 8: Low Gamma-Ray Emission (Solar Max, Oct. 2024)

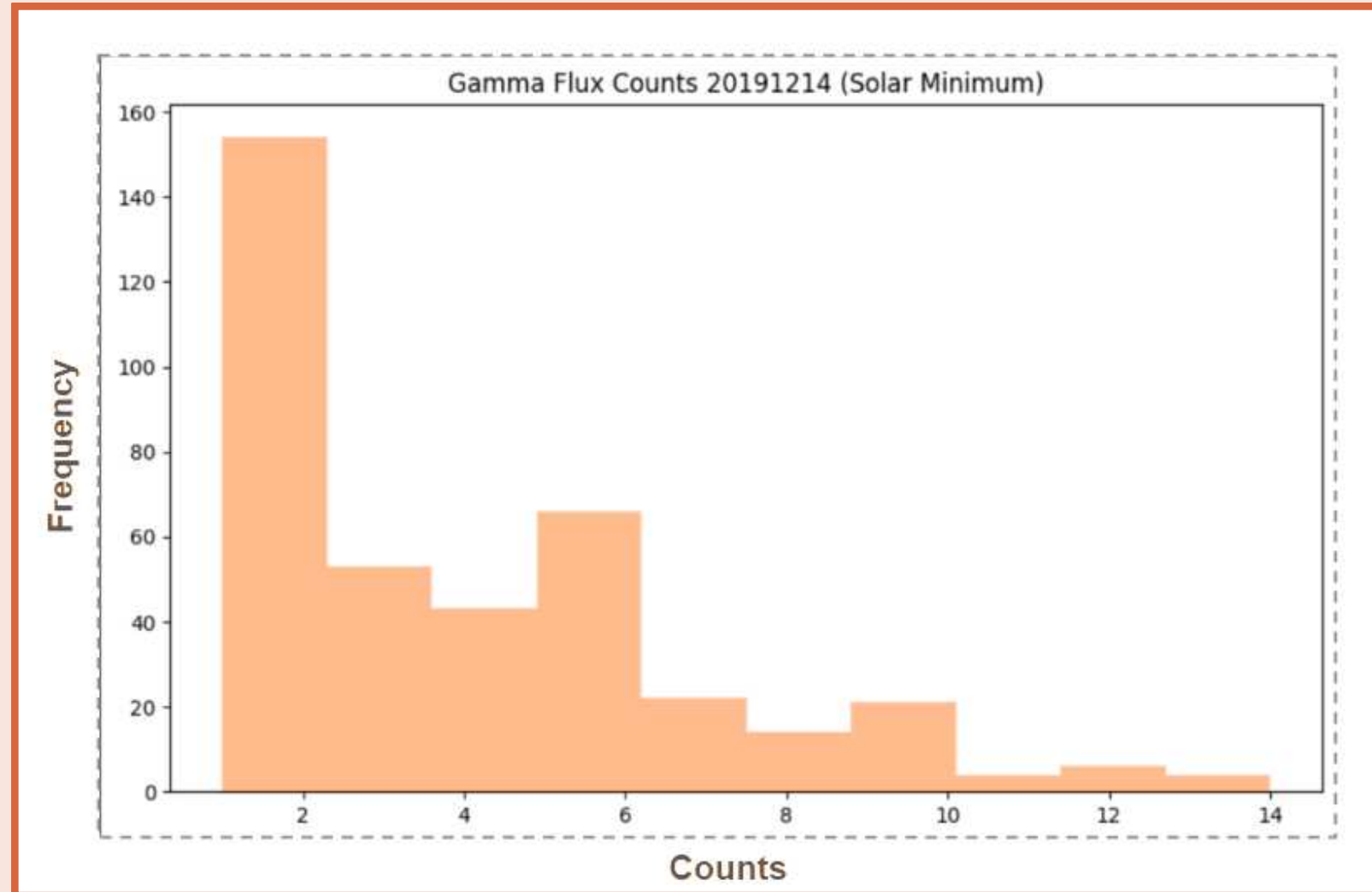


Figure 9: High Gamma-Ray Emission (Solar Min, Dec. 2019)

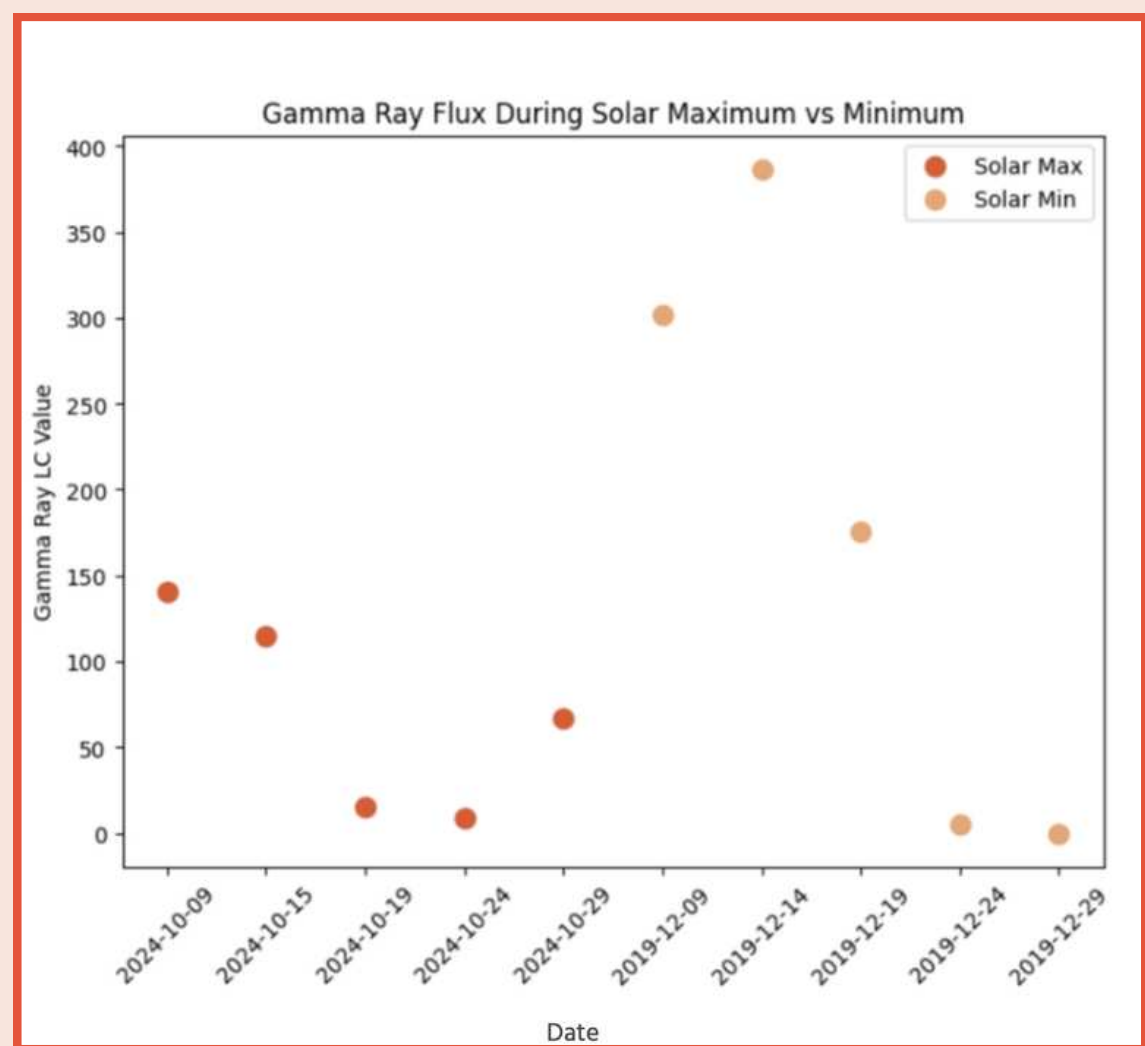


Figure 10: GRF During Different Stages of the Solar Cycle

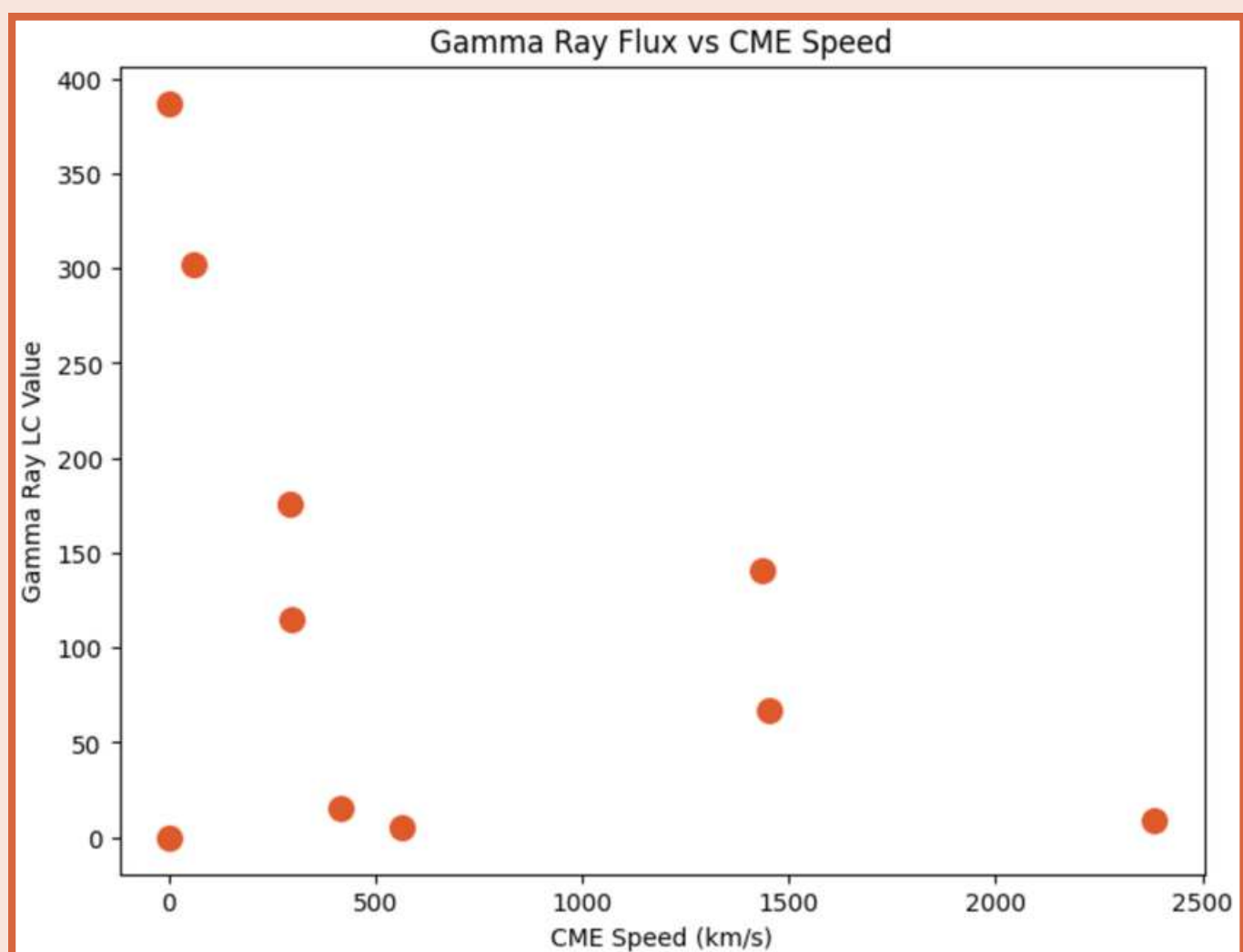


Figure 11: Comparison of GRF and CME Speed

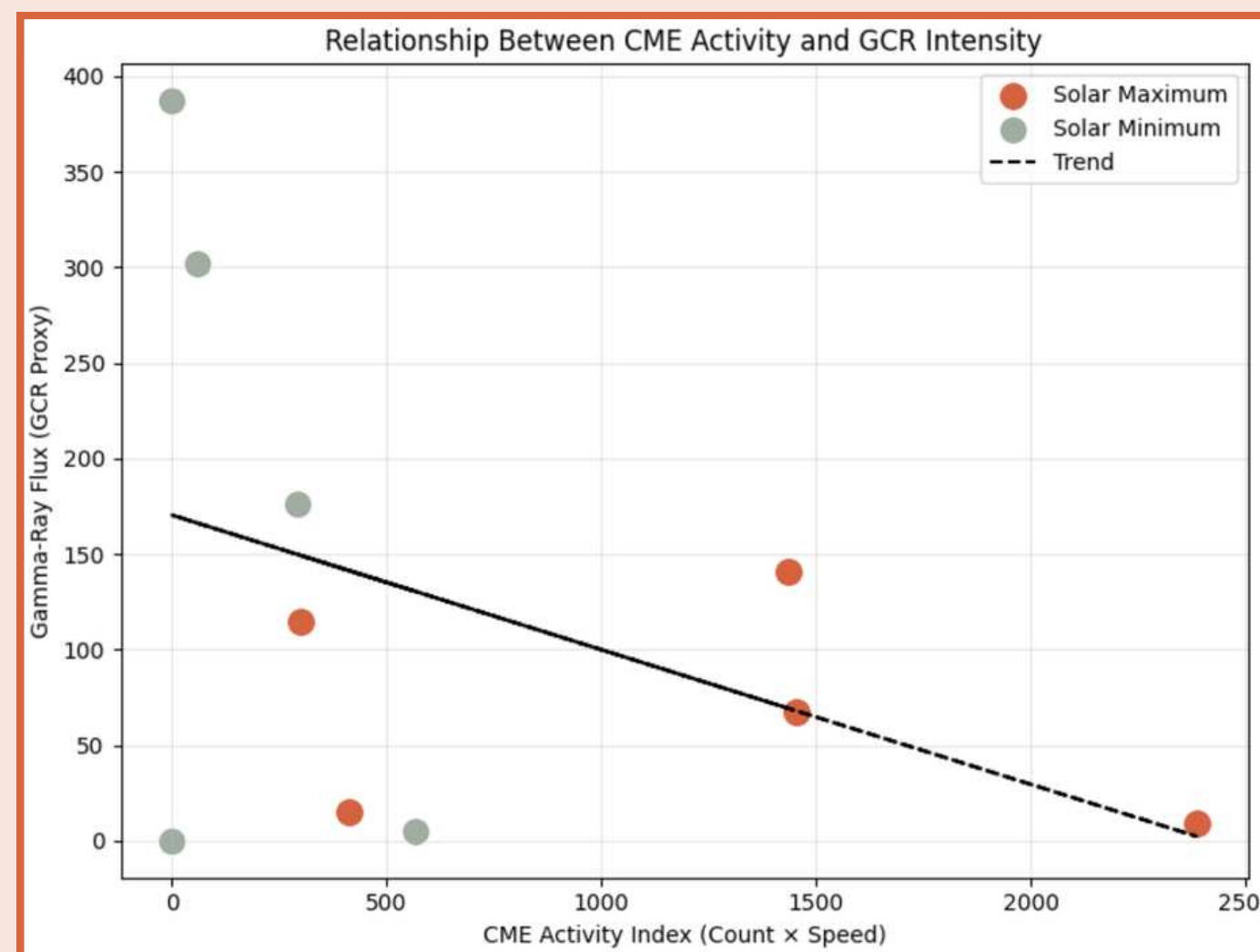


Figure 12: Correlation between CME Activity + GCRs

Lower flux during solar max; higher flux during solar min

Greater flux, lower speed; lower flux, higher speed

DISCUSSION

Gamma-ray flux values were generally higher during solar minimum and lower during solar maximum, supporting the hypothesis that increased solar magnetic activity reduces Galactic Cosmic Ray (GCR) intensity near Earth. This is consistent with the concept of solar modulation, where a stronger heliospheric magnetic field during solar maximum deflects incoming cosmic rays.

Additionally, an inverse relationship was observed between CME activity and gamma-ray flux. Periods with higher CME speeds corresponded to lower gamma-ray flux values, suggesting that increased solar activity further suppresses GCR penetration. However, the correlation was not perfectly linear, likely due to the limited sample size and natural variability in solar events.

Overall, the results support the idea that solar activity plays a significant role in regulating high-energy particle interactions near Earth, though more data is needed to confirm the strength of this relationship.

NEXT STEPS

- Increase sample size by analyzing more dates across multiple solar cycles
- Incorporate continuous datasets instead of single-day samples for improved accuracy
- Use additional solar activity indicators (e.g., sunspot number, solar wind data from NOAA)
- Refine gamma-ray flux calculations using full spectral data instead of light curve averages
- Perform stronger statistical analysis (e.g., correlation coefficients, regression significance)
- Compare results with published models of solar modulation

REFERENCES

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